

TYPES OF ALGORITHMS

Single Stock Algorithms: Single stock algorithms interact with the market based on user specified settings and will take advantage of favorable market conditions only when it is in the best interest of the order and the investor. Single stock algorithms are independent of one another while trading in the market and make decisions based solely on how those decisions will affect the individual order.

VWAP: Volume weighted average price. These algorithms participate in proportion with the intraday volume curve. If 5% of the day's volume trades in any specified period then the VWAP algorithm will transact 5% of the order in that period. The intraday volume profile used to follow a U-shaped pattern with more volume traded at the open and close than mid-day. But recently, intraday volume profiles have become more back-loaded and resemble more of a J-shaped pattern than U-shaped pattern. A VWAP strategy is a static strategy and will remain constant throughout the day.

TWAP: Time weighted average price. These algorithms execute orders following a constant participation rate through the entire day. A full day order will trade approximately 1/390th of the order in each 1 minute bucket (there are 390 minutes in the trading day in the US). It is important to note that many TWAP algorithms do not participate with volume in the opening and closing auctions since there is no mathematical method to determine the quantity of shares to enter into these auctions. In *Optimal Trading Strategies*, the TWAP curve was referred to as the uniform distribution or uniform strategy and was used for comparison purposes. A TWAP strategy is a static strategy and will remain constant throughout the day.

Volume: These strategies are referred to volume inline, percentage of volume (POV), or participation rate algorithms. These algorithms participate with market volume at a pre-specified rate such as 20% and will continue to trade until the entire order is completed. The algorithms will trade more shares in times of higher liquidity and fewer shares in times of lower liquidity, and thus react to market conditions (at least to changing volume profiles). One drawback to these volume strategies is that they do not guarantee completion of the order by the end of the time horizon. For example, if we are trading an order that comprises 20% of the day's volume at a $POV = 20\%$ rate but the actual volume on the day is only half of its normal volume, the order would not complete by the end of the day. As a safety around potential uncompleted orders, some brokers have offered a parameter to ensure completion by the end of the period. This parameter serves as a minimum POV rate and adjusts in real-time to ensure order completion by the designated end time.